Early Triassic Carbon Cycle Reconstruction and Dynamical System Analysis

1. Project Overview

1.1 Scientific Background

The Early-Middle Triassic (~252-242 Ma) represents a critical recovery interval following the end-Permian mass extinction, marked by severe environmental perturbations. Large-scale volcanic activity, such as the Siberian Traps, released massive amounts of CO_2 into the atmosphere, triggering greenhouse warming, ocean anoxia, and acidification.Geological records indicate pronounced periodic climate variability during this time, with nonlinear responses in the carbon cycle. However, the underlying mechanisms remain unclear.This project uses carbonate lithium isotope (δ^7Li) data as an independent geochemical constraint, coupled with a multi-process carbon cycle model, to reconstruct atmospheric CO_2 and global temperature. The focus is on the coupling between volcanic activity, continental weathering (including reverse weathering), and climate stability under extreme conditions.

1.2 Objectives

1. Reconstruct atmospheric CO_2 and temperature anomalies (242-252 Ma) from carbonate δ^7Li data using detrital and fractionation corrections.

2. Develop a nonlinear dynamical carbon cycle model including volcanic input, weathering sinks, and ocean anoxia feedbacks to identify possible bistability and tipping behavior.

3. Quantify uncertainty using Monte Carlo simulations to assess parameter sensitivity and improve robustness.

2. Data and Preprocessing

2.1 Data

Data are derived from 19 carbonate samples from the Xiejiachao section (Sichuan Basin), including age, δ^7Li, and elemental composition.

Reference state: $\delta^7Li_{ref} = 8.1‰$， $CO_{2\,ref} = 2400$ ppm

2.2 Detrital Correction and Seawater δ^7Li

Bulk-rock samples may contain detrital contamination. Aluminum (Al) is used as a proxy to estimate detrital fraction.

Corrected carbonate δ^7Li is calculated using mass balance, followed by fractionation correction:

$f_{det}=\frac{Al_{sample}}{Al_{det}}$, $Al_{det}=8\%$

If the calculated $f_{det}\geq 1.0$, $f_{det}=0.99$ for subsequent calculations.

The δ^7Li value of the carbonate component is calculated using mass balance:

$$\delta^7Li_{carb}=\frac{\delta^7Li_{whole}-f_{det}\cdot\ \delta^7Li_{det}}{1-f_{det}}$$

Where δ^7Li_{whole} is the bulk-rock δ^7Li value, and $f_{det}\cdot\ \delta^7Li_{det}=1.0‰$ is the typical δ^7Li value of terrigenous detritus.

Lithium isotope fractionation occurs during carbonate precipitation. Based on existing studies, a fractionation factor ($\Delta_{carb\text{-}sw}=4.0‰$) is set (the offset of δ^7Li in carbonate relative to seawater). The paleo-seawater δ^7Li value is calculated as:

$$\delta^7Li_{sw}=\delta^7Li_{carb}+\Delta_{carb-sw}$$

2.3 Extrapolation (242–244 Ma)

The measured samples only cover the 244-251.5 Ma interval. To reconstruct the full seawater δ^7Li evolution from 242 to 252 Ma, extrapolation is performed for the 242-244 Ma interval. Considering the gradual closure of the sedimentary system after the Early Triassic transgression and the return of lithium isotope composition to regional steady state, an "exponential decay + plateau" model is used:

$$\delta^7Li_{sw}(t)=Li_{eq}+(Li_{peak}-Li_{eq})\cdot\ exp(-\frac{244-t}{\tau})$$

Parameter definitions and ranges:

Li_{peak} : seawater δ^7Li value at 244 Ma, adjusted with detrital and fractionation correction

Li_{eq} : regional steady-state background seawater δ^7Li value, ranging from 10–12‰

τ : recovery time constant of lithium isotope composition, representing the system's return rate to steady state after perturbation, ranging from 0.5–1.5 Myr.

3. Reconstruction Model (Zero-Dimensional Carbon Cycle)

3.1 Lithium Logarithmic Deviation

To quantify the deviation of paleo-seawater δ^7Li from the reference state, a logarithmic deviation parameter $x(t)$ is defined as the core input signal for weathering flux calculations:

$$x(t)=ln(\frac{\delta^7Li_{sw}}{8.1})$$

3.2 Volcanic Input Flux $F_{vol}(t)$

Volcanic activity was the primary source of atmospheric CO_2 during the Early Triassic. The volcanic input flux $F_{vol}(t)$ is modeled as a background flux with superimposed volcanic pulses, each described by a Gaussian function:

$$F_{vol}(t) = F_0 + \sum_i A_i exp[-\frac{(t-t_i)^2}{2\sigma^2}]$$

Three major volcanic events during the Early Triassic are identified with the following parameters, with a background value F_0 = 0.0002 representing stable CO_2 input during non-volcanic periods.

Center Time t_i (Ma)	Width σ_i (Myr)	Amplitude A_i	Meaning
251.4	0.15	1.0	Late Griesbachian volcanism
249.56	0.08	0.85	SSB volcanism
247.2	0.08	0.6	OAB volcanism

3.3 Weathering Consumption Flux $F_w(t)$

Continental weathering is the main sink for atmospheric CO_2, while reverse weathering (clay formation) releases CO_2, together forming the source-sink balance of the carbon cycle. The weathering consumption flux $F_w(t)$ is divided into silicate weathering flux F_{sil} and clay formation/reverse weathering flux F_{clay}, with reverse weathering regulated by a positive temperature feedback:

$$\eta_{rev} = \eta_{rev0}\,[1 + k_T \max(0, \Delta T)]$$

$$\eta_{\mathrm{eff}} = \eta_{pos} - \eta_{rev}$$

$$F_{sil} = F_{s0}\, exp\,(E_{sil}\Delta T)$$

$$F_{clay} = exp\,(\eta_{\mathrm{eff}}\, x)$$

$$F_w = F_{sil} + F_{clay}$$

Typical parameter values and meanings: η_{pos} =0.15 (positive weathering sensitivity, reflecting the response intensity of silicate weathering to lithium isotopes); η_{rev}=1.0 (baseline reverse weathering sensitivity); k_T=0.5 （(temperature feedback coefficient, regulating reverse weathering intensity by temperature); F_{s0} =1.2 (baseline silicate weathering flux); E_{sil} =0.8 (silicate weathering temperature coefficient, reflecting temperature enhancement of silicate weathering). $\max(0, \Delta T)$ indicates that reverse weathering is enhanced only when the temperature anomaly is positive.

3.4 Source-to-Sink Ratio and CO_2 Concentration

The source-to-sink ratio $r(t)$ is defined as the ratio of volcanic input flux to weathering consumption flux:

$$r = \frac{F_{vol}}{F_w}$$

This ratio directly determines the trend of atmospheric CO_2 concentration. Using the reference CO_2 concentration, atmospheric CO_2 is calculated as:

$$CO_2(t) = 2400 \cdot r(t)^{\alpha(t)} \cdot osc(t)$$

$\alpha(t)$: nonlinear response coefficient, increasing linearly from 0.05 to 0.16 during 249–248 Ma, reflecting the gradual increase in carbon cycle sensitivity.

$osc(t)$: orbital-scale oscillation factor, modeling orbital forcing (e.g., Milankovitch cycles) on the carbon cycle. A smooth window function activates oscillations within the measured interval and smoothly dampens them outside to avoid boundary artifacts:

$$osc(t) = 1 + \beta sin\,(2\pi \frac{T}{t - t_0})\, w(t)$$

Where $\beta = 0.15$ (oscillation amplitude), $T = 3.0$ Myr (oscillation period), t_0=252 Myr (oscillation start time), and $w(t)$ is a cubic smoothstep function with a transition width of 1 Myr.

3.5 Temperature Anomaly

Based on the greenhouse effect, changes in atmospheric CO_2 concentration directly affect global temperature, described by a logarithmic relationship:

$$\Delta T = \lambda \ln\left(\frac{CO_2}{2400}\right), \lambda = 4.0°C$$

3.6 Solution Procedure

Since the reverse weathering sensitivity η_{rev} depends on the temperature anomaly ΔT, and ΔT depends on CO_2 concentration, which in turn depends on the weathering consumption F_w, a coupled relationship is formed. To solve this, a fixed-point iteration method is used at each time step, with a maximum of 30 iterations and a convergence tolerance of 10^{-4}.

4. Dynamical System Model (Three-Variable ODE)

To further explore the nonlinear dynamics of the Early Triassic carbon cycle and climate system, the reconstructed volcanic flux $V(t) = F_{vol}(t)$ is used as an external forcing to construct a three-variable nonlinear ordinary differential equation (ODE) system. This system analyzes the critical transition mechanisms of the climate system under coupled volcanism-

weathering-anoxia feedbacks.

4.1 State Variables

Variable	Definition	Physical Meaning
c	$CO_2/2400$	Dimensionless CO_2 carbon reservoir, relative to reference state
T	Dimensionless temperature	Temperature state, equilibrates with ln c
a	Anoxia index	Ocean anoxia level, ranging from 0 (fully oxic) to 1 (fully anoxic)

4.2 Governing Equations

$$\frac{dc}{dt} = V(t) - c^{\beta} + \alpha \cdot \frac{a^n}{K^n + a^n}$$

$$\frac{dT}{dt} = lnc - T$$

$$\frac{da}{dt} = \sigma \cdot \frac{T^m}{H^m + T^m} - a$$

1. Carbon equation: Volcanic injection $V(t)$ 使 increases the carbon reservoir; weathering acts as a sink, consuming carbon as c^{β}; ocean anoxia releases CO_2 to the atmosphere via a Hill function. $(\frac{a^n}{K^n+a^n})$, forming a positive feedback, with a as the maximum CO_2 release intensity from anoxia.

2. Temperature equation: Temperature state T relaxes toward the equilibrium temperature ln c reflecting the lag in temperature response to CO_2 changes.

3. Anoxia equation: Temperature drives ocean anoxia via a Hill function $\frac{T^m}{H^m+T^m}$ while anoxia naturally decays. σ is the maximum anoxia intensity driven by temperature.

4.3 Parameters

Parameter	Value	Meaning
β	0.3	Weathering sink nonlinear exponent
α	5.0	Maximum CO_2 release from anoxia
σ	5.0	Maximum anoxia intensity driven by temperature
n	4	Hill coefficient for anoxia feedback

m	4	Hill coefficient for temperature-driven anoxia
K	0.5	Anoxia feedback threshold
H	0.5	Temperature-driven anoxia threshold

4.4 Steady State and Bifurcation Analysis

Setting the time derivatives to zero yields the steady-state solutions c^* ; a^* , related to external forcing V.

$$a^* = \sigma \cdot \frac{(\ln c^*)^m}{H^m + (\ln c^*)^m}, \quad V = (c^*)^\beta - \alpha \frac{(a^*)^n}{K^n + (a^*)^n}$$

Multiple steady-state roots are found via multi-initial-value scanning to generate bifurcation diagrams (V vs. c^*). An S-shaped curve indicates the existence of saddle-node bifurcations and bistability, where the climate system may undergo abrupt transitions under perturbations.

5. Code Structure and Execution

5.1 Dependencies

numpy: numerical calculations, array operations

scipy: numerical solvers (fsolve for fixed-point iteration, solve_ivp for ODE, interp1d for interpolation)

matplotlib: plotting (panel figures, bifurcation diagrams)

pandas: data processing, table generation, Excel I/O

openpyxl: Excel file export and editing

5.2 File Structure

lithium-isotope-model.py # Main program

README.md # Project documentation

5.3 Execution

1. Install dependencies: `pip install numpy scipy matplotlib pandas openpyxl`

2. Run the program: `python lithium-isotope-model.py`

6. Output and Figures

The 3×3 panel figure is generated, including:

1. Lithium isotope correction (raw vs. corrected seawater δ^7Li)

2. Volcanic input flux with labeled pulses

3. Baseline CO_2 concentration with reference line

4. Temperature anomaly comparison (baseline vs. theoretical scenario)

5. Volcanic contribution to temperature anomaly

6. CO_2 uncertainty (percentile bands from Monte Carlo)

7. Temperature uncertainty (percentile bands)

8. Weathering decomposition (silicate vs. clay formation fluxes)

9. Source-to-sink ratio

Solid lines indicate measured intervals (≥ 244 Ma), dashed lines indicate extrapolated intervals (< 244 Ma). A vertical dashed line marks 244 Ma. Figure size: 34×18 cm, font: Calibri 15.

7. Parameter Settings and Uncertainty Quantification

7.1 Monte Carlo Sampling Ranges

500 Monte Carlo simulations are performed with uniform random sampling over key parameters, covering correction, volcanic, weathering, oscillation, climate, and extrapolation categories.

7.2 Fixed Parameters

Timing and endpoints of α(t) transition (249-248 Ma, 0.05 to 0.16)

Window function transition width (1 Myr)

Center times of volcanic events (251.4, 249.56, 247.2 Ma)。

8. Environment Requirements

Python 3.8–3.10

Dependencies: numpy, scipy, matplotlib, pandas, openpyxl